Relativistic Quantum Mechanics III: Electrons in an External Field

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1 Dirac Equation in an External Field

1.1 Dirac Equation in an External Electromagnetic Field

The wave equation of a free particle is determined by the requirement of space-time symmetry, but physical processes involving particles depend on their interaction properties. In QED, we discuss the interaction between electrons (spin $\frac{1}{2}$ particles) and electromagnetic (EM) fields. In this note, we restrict ourselves on the "single-particle theory", where **the number of particles doesn't change, and the electromagnetic filed is then treated as a "given" external field.** The validity of this formalism is obviously limited by the conditions where a field can be viewed as "given". Conditions arising from the radiative corrections also impose limitations.

In classical electrodynamics, an EM field is described by a 4-potential. In relativistic units ($\hbar = 1,\ c = 1$),

$$A^\mu = (\varphi, \boldsymbol{A}) \tag{1.1}$$

The canonical momentum of an electron in this EM field is

$$p^\mu = P^\mu + eA^\mu \tag{1.2}$$

where P^μ is the kinetic momentum, and e is the electronic charge, which is negative ($e = -|e|$). The derivation of the relativistic quantum mechanical equation of motion for electrons in an external field follows the same procedure as in non-relativistic quantum mechanics, i.e., we first replace the canonical momentum by the operator

$$\hat{p}^\mu = i\partial^\mu = i\left(\frac{\partial}{\partial t}, -\nabla\right) \tag{1.3}$$

and then replace the kinetic momentum operator $\hat{p}^\mu$ in the Dirac equation for free particles, $(\gamma\hat{p} - m)\psi = 0$, by the kinetic momentum operator in the field, $\hat{P}^\mu = \hat{p}^\mu - eA^\mu$. We then arrived at the **Dirac equation for an electron in an external EM field**,

$$\boxed{[\gamma(\hat{p} - eA) - m]\psi = 0} \tag{1.4}$$

The Hamiltonian operator is given by

$$\boxed{\hat{H} = \boldsymbol{\alpha} \cdot (\hat{\boldsymbol{p}} - e\boldsymbol{A}) + \beta m + e\varphi} \tag{1.5}$$

The EM field is invariant under a gauge transformation,

$$A^\mu \to A^\mu - \partial^\mu \chi \equiv A^\mu + i\hat{p}^\mu \chi \tag{1.6}$$

where $\chi(\boldsymbol{r}, t)$ is an arbitrary function of space and time. The Dirac equation should also be invariant under such transformation. Substituting $A^\mu = A'^\mu + i\hat{p}^\mu\chi$ in Eq. (1.4), and multiplying $e^{ie\chi}$ on both sides of the equation, we have

$$\gamma[(\hat{p}\psi)e^{ie\chi} - eA'\psi e^{ie\chi} + ie(\hat{p}\chi)\psi e^{ie\chi}] = m\psi e^{ie\chi} \tag{1.7}$$

Since $(\hat{p}\psi)e^{ie\chi} + ie(\hat{p}\chi)\psi e^{ie\chi} = \hat{p}(\psi e^{ie\chi})$, the equation can be simplified as

$$[\gamma(\hat{p} - eA') - m]\psi' = 0 \tag{1.8}$$

where $\psi' = \psi e^{ie\chi}$. **Performing a gauge transformation on the EM field leads to a change of the phase factor in the particle's wave function. The Dirac equation is invariant under gauge transformations.**

Next, we apply the charge conjugation operation to Eq. (1.4). To do so, we first take the Dirac conjugation of Eq. (1.4) and transpose,

$$[\gamma^{\mathrm{T}}(\hat{p} + eA) + m]\bar{\psi}^{\mathrm{T}} = 0 \tag{1.9}$$

Then we multiply $U_C = \gamma^2\gamma^0$ from left, and use the commutation relation between U_C and γ^μ:

$$U_C(\gamma^\mu)^{\mathrm{T}} = -\gamma^\mu U_C \tag{1.10}$$

we have

$$\boxed{[\gamma(\hat{p} + eA) - m]\psi^C = 0} \tag{1.11}$$

where $\psi^C = \hat{C}\psi = U_C\bar{\psi}^{\mathrm{T}}$ is the wave function of the particle after charge conjugation operation, i.e., the wave function of the antiparticle. **The antiparticle wave function satisfies an equation which differs from the original equation by a change in sign of the charge. The signs of the electron and position charges are thus opposite.**

Just as in non-relativistic theory, the solutions of the Dirac equation in an external field may include states with continuous spectrum and discrete spectrum. The former corresponds to free particle states with infinite motion, whereas the later corresponds to bound states. Since the eigenvalues of the free particle Hamiltonian is $\pm\sqrt{\boldsymbol{p}^2 + m^2}$, the continuous spectrum should have the energy range $\varepsilon \geq m$ and $\varepsilon \leq -m$, and the discrete spectrum should have the energy range of $-m < \varepsilon < m$.

In the free particle case, solutions with positive energies correspond to particles and solutions with negative energies correspond to antiparticles. For the case of particles in an external field, however, the particle solutions may also possess negative energies when the potential well becomes deeper, and *vice versa*. Therefore, **the bound particle states should be regarded as those whose energies approach $\varepsilon = m$ limit when the potential is gradually removed, and the**

bound antiparticle states should be regarded as those whose energies approach $\varepsilon = -m$ **limit when the potential is gradually removed.**

Let $\psi_n^{(+)}$ and $\psi_n^{(-)}$ be the appropriately normalized eigenfunctions of the Hamiltonian Eq. (1.5), with corresponding eigenvalues $\varepsilon_n^{(+)}$ and $-\varepsilon_n^{(-)}$, the field operators can then be expressed as

$$\boxed{\hat{\psi} = \sum_n \left(\hat{a}_n \psi_n^{(+)} e^{-i\varepsilon_n^{(+)} t} + \hat{b}_n^\dagger \psi_n^{(-)} e^{i\varepsilon_n^{(-)} t} \right)} \tag{1.12}$$

$$\boxed{\hat{\bar{\psi}} = \sum_n \left(\hat{a}_n^\dagger \bar{\psi}_n^{(+)} e^{i\varepsilon_n^{(+)} t} + \hat{b}_n \bar{\psi}_n^{(-)} e^{-i\varepsilon_n^{(-)} t} \right)} \tag{1.13}$$

1.2 Second-Order Equation

For a free particle, we formulated a second-order equation, $(\hat{p}^2 - m^2)\psi = 0$, for ψ. For a particle in an external field, we can also transform Eq. (1.4) into a second-order equation by applying the operator $\gamma(\hat{p} - eA) + m$,

$$[\gamma^\mu \gamma^\nu (\hat{p}_\mu - eA_\mu)(\hat{p}_\nu - eA_\nu) - m^2]\psi = 0 \tag{1.14}$$

The product $\gamma^\mu\gamma^\nu$ can be written in the form

$$\gamma^\mu\gamma^\nu = \frac{1}{2}(\gamma^\mu\gamma^\nu + \gamma^\nu\gamma^\mu) + \frac{1}{2}(\gamma^\mu\gamma^\nu - \gamma^\nu\gamma^\mu) \equiv g^{\mu\nu} + \sigma^{\mu\nu} \tag{1.15}$$

where $g^{\mu\nu}$ is the metric tensor, and $\sigma^{\mu\nu}$ is the **antisymmetrical matrix 4-tensor**,

$$\boxed{\sigma^{\mu\nu} = \frac{1}{2}(\gamma^\mu\gamma^\nu - \gamma^\nu\gamma^\mu) = (\boldsymbol{\alpha}, i\boldsymbol{\Sigma})} \tag{1.16}$$

and

$$\boldsymbol{\Sigma} = \begin{pmatrix} \boldsymbol{\sigma} & 0 \\ 0 & \boldsymbol{\sigma} \end{pmatrix} \tag{1.17}$$

We can then write second-order equation as

$$[(\hat{p} - eA)^2 - m^2]\psi + \sigma^{\mu\nu}(\hat{p}_\mu - eA_\mu)(\hat{p}_\nu - eA_\nu)\psi = 0 \tag{1.18}$$

Since $\sigma^{\mu\nu}$ is anti symmetric, we have

$$\sigma^{\mu\nu}(\hat{p}_\mu - eA_\mu)(\hat{p}_\nu - eA_\nu) = -\sigma^{\nu\mu}(\hat{p}_\mu - eA_\mu)(\hat{p}_\nu - eA_\nu) = -\sigma^{\mu\nu}(\hat{p}_\nu - eA_\nu)(\hat{p}_\mu - eA_\mu) \tag{1.19}$$

Thus

$$\begin{aligned}
\sigma^{\mu\nu}(\hat{p}_\mu - eA_\mu)(\hat{p}_\nu - eA_\nu)\psi &= \frac{1}{2}\sigma^{\mu\nu}[(\hat{p}_\mu - eA_\mu)(\hat{p}_\nu - eA_\nu) - (\hat{p}_\nu - eA_\nu)(\hat{p}_\mu - eA_\mu)]\psi \\
&= \frac{1}{2}e\sigma^{\mu\nu}[-A_\mu\hat{p}_\nu\psi - \hat{p}_\mu(A_\nu\psi) + A_\nu\hat{p}_\mu\psi + \hat{p}_\nu(A_\mu\psi)] \\
&= \frac{1}{2}e\sigma^{\mu\nu}(-\hat{p}_\mu A_\nu + \hat{p}_\nu A_\mu)\psi \\
&= -\frac{1}{2}ie\sigma^{\mu\nu}(\partial_\mu A_\nu - \partial_\nu A_\mu)\psi \\
&= -\frac{1}{2}ie\sigma^{\mu\nu}F_{\mu\nu}\psi
\end{aligned} \tag{1.20}$$

where

$$F^{\mu\nu} = \partial^\mu A^\nu - \partial^\nu A^\mu = (-\boldsymbol{E}, \boldsymbol{B}) \tag{1.21}$$

is the EM field tensor. We arrived at the second-order equation

$$\boxed{\left[(\hat{p} - eA)^2 - m^2 - \frac{1}{2} ieF_{\mu\nu}\sigma^{\mu\nu}\right]\psi = 0} \tag{1.22}$$

Since the contraction of two antisymmetrical 4-tensors $A^{\mu\nu} = (\boldsymbol{p}, \boldsymbol{a})$, $B^{\mu\nu} = (\boldsymbol{q}, \boldsymbol{b})$ can be written in terms as its component as

$$A_{\mu\nu}B^{\mu\nu} = -2\boldsymbol{p}\cdot\boldsymbol{q} + 2\boldsymbol{a}\cdot\boldsymbol{b} \tag{1.23}$$

we have

$$F_{\mu\nu}\sigma^{\mu\nu} = 2\boldsymbol{\alpha}\cdot\boldsymbol{E} + 2i\boldsymbol{\Sigma}\cdot\boldsymbol{B} \tag{1.24}$$

and

$$\left[(\hat{p} - eA)^2 - m^2 + e\boldsymbol{\Sigma}\cdot\boldsymbol{B} - ie\boldsymbol{\alpha}\cdot\boldsymbol{E}\right]\psi = 0 \tag{1.25}$$

The involvement of the fields $\boldsymbol{E}$ and $\boldsymbol{B}$ is due to the spin of the particle.

Note that the second-order equation will have "redundant solutions". Those are the solutions of the equation $[\gamma(\hat{p} - eA) + m]\phi = 0$, but not the solutions of Eq. (1.4). However, for any solution ϕ of the second-order equation, $\psi = [\gamma(\hat{p} - eA) + m]\phi$ will be a solution of the correct first-order Dirac equation. It also should be emphasized that in deriving the relativistic wave equations of particles in the field, the replacement of $\hat{p} \to \hat{p} - eA$ must be made in the first order equation, otherwise the term $-\frac{1}{2}ieF_{\mu\nu}\sigma^{\mu\nu}$ will be missing in Eq. (1.22).

1.3 Ordinary Units

We can also express the first-order Dirac equation and the second-order equation in SI or Gaussian units. These expressions are especially useful when we expanding the equations in powers of $1/c$.

For the expressions in the SI units, We make the following substitutions in the equations,

$$\begin{aligned} m &\to mc \\ \hat{p}^\mu &= i\hbar\partial^\mu = i\hbar\left(\frac{1}{c}\frac{\partial}{\partial t}, -\nabla\right) \\ A^\mu &= \left(\frac{\varphi}{c}, \boldsymbol{A}\right) \\ F^{\mu\nu} &= \left(-\frac{\boldsymbol{E}}{c}, \boldsymbol{B}\right) \end{aligned} \tag{1.26}$$

Thus

$$\hat{p}^\mu - eA^\mu = \left(\frac{i\hbar}{c}\frac{\partial}{\partial t} - \frac{e\varphi}{c}, -i\hbar\nabla - e\boldsymbol{A}\right) \tag{1.27}$$

The Dirac equation becomes

$$\left[\left(\frac{i\hbar}{c}\frac{\partial}{\partial t} - \frac{e\varphi}{c}\right) - \boldsymbol{\alpha}\cdot(-i\hbar\nabla - e\boldsymbol{A}) - \beta mc\right]\psi = 0 \tag{1.28}$$

or

$$\boxed{i\hbar\frac{\partial}{\partial t}\psi = \left[c\boldsymbol{\alpha}\cdot(-i\hbar\nabla - e\boldsymbol{A}) + \beta mc^2 + e\varphi\right]\psi} \tag{1.29}$$

The Hamiltonian operator in SI units is thus

$$\boxed{\hat{H} = c\boldsymbol{\alpha}\cdot(-i\hbar\nabla - e\boldsymbol{A}) - \beta mc^2 + e\varphi} \tag{1.30}$$

And the second-order equation in SI units is

$$\boxed{\left[\left(\frac{i\hbar}{c}\frac{\partial}{\partial t} - \frac{e\varphi}{c}\right)^2 - (i\hbar\nabla + e\boldsymbol{A})^2 - m^2c^2 + e\hbar\boldsymbol{\Sigma}\cdot\boldsymbol{B} - i\frac{e\hbar}{c}\boldsymbol{\alpha}\cdot\boldsymbol{E}\right]\psi = 0} \tag{1.31}$$

For the expressions in the Gaussian units, We make the following substitution instead,

$$\begin{aligned} m &\to mc \\ e &\to \frac{e}{c} \\ \hat{p}^\mu &= i\hbar\partial^\mu = i\hbar\left(\frac{1}{c}\frac{\partial}{\partial t}, -\nabla\right) \\ A^\mu &= (\varphi, \boldsymbol{A}) \\ F^{\mu\nu} &= (-\boldsymbol{E}, \boldsymbol{B}) \end{aligned} \tag{1.32}$$

Thus

$$\hat{p}^\mu - \frac{e}{c}A^\mu = \left(\frac{i\hbar}{c}\frac{\partial}{\partial t} - \frac{e\varphi}{c}, -i\hbar\nabla - \frac{e}{c}\boldsymbol{A}\right) \tag{1.33}$$

The Dirac equation becomes

$$\left[\left(\frac{i\hbar}{c}\frac{\partial}{\partial t} - \frac{e\varphi}{c}\right) - \boldsymbol{\alpha}\cdot\left(-i\hbar\nabla - \frac{e}{c}\boldsymbol{A}\right) - \beta mc\right]\psi = 0 \tag{1.34}$$

or

$$i\hbar\frac{\partial}{\partial t}\psi = \left[c\boldsymbol{\alpha}\cdot\left(-i\hbar\nabla - \frac{e}{c}\boldsymbol{A}\right) + \beta mc^2 + e\varphi\right]\psi \tag{1.35}$$

The Hamiltonian operator in Gaussian units is thus

$$\hat{H} = c\boldsymbol{\alpha}\cdot\left(-i\hbar\nabla - \frac{e}{c}\boldsymbol{A}\right) - \beta mc^2 + e\varphi \tag{1.36}$$

And the second-order equation in Gaussian units is

$$\left[\left(\frac{i\hbar}{c}\frac{\partial}{\partial t} - \frac{e\varphi}{c}\right)^2 - \left(i\hbar\nabla + \frac{e}{c}\boldsymbol{A}\right)^2 - m^2c^2 + \frac{e\hbar}{c}\boldsymbol{\Sigma}\cdot\boldsymbol{B} - i\frac{e\hbar}{c}\boldsymbol{\alpha}\cdot\boldsymbol{E}\right]\psi = 0 \tag{1.37}$$

The only difference between the equations in Gaussian and SI units is the additional $1/c$ factor in front of the vector potential $\boldsymbol{A}$ and the magnetic field $\boldsymbol{B}$ in Gaussian units.

2 Non-Relativistic Limit

In the standard representation, two of the components (χ) in the bispinor wave function of the free particle become zero in its rest frame,

$$\psi = \begin{pmatrix} \phi \\ \chi \end{pmatrix} \to \begin{pmatrix} \phi \\ 0 \end{pmatrix}, \quad \text{when } \boldsymbol{p} \to 0 \tag{2.1}$$

Therefore, in the non-relativistic limit, where the velocity of the particle $v \ll c$, $\chi \ll \phi$, we can obtain an approximate equation involving only the two-component quantity ϕ. This can be done by a formal expansion of the wave function in powers of $1/c$.

In SI units, the Dirac equation is

$$i\hbar \frac{\partial}{\partial t}\psi = \left[c\boldsymbol{\alpha} \cdot (\hat{\boldsymbol{p}} - e\boldsymbol{A}) + \beta mc^2 + e\varphi\right] \psi \tag{2.2}$$

The relativistic energy of the particle includes its rest energy mc^2, which should be excluded in the non-relativistic approximation. The shift of energy leads to a change in the phase factor of ψ. Let

$$\psi' = \psi e^{i\frac{mc^2 t}{\hbar}} \tag{2.3}$$

then

$$\left(i\hbar\frac{\partial}{\partial t} + mc^2\right) \psi' = \left[c\boldsymbol{\alpha} \cdot (\hat{\boldsymbol{p}} - e\boldsymbol{A}) + \beta mc^2 + e\varphi\right] \psi' \tag{2.4}$$

Substituting the definition of $\boldsymbol{\alpha}$ and β matrices in standard representation,

$$\boldsymbol{\alpha} = \begin{pmatrix} 0 & \boldsymbol{\sigma} \\ \boldsymbol{\sigma} & 0 \end{pmatrix}, \quad \beta = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \tag{2.5}$$

we arrived at the coupled equations for two-component quantities ϕ' and χ',

$$\boxed{\begin{aligned} \left(i\hbar\frac{\partial}{\partial t} - e\varphi\right) \phi' &= c\boldsymbol{\sigma} \cdot (\hat{\boldsymbol{p}} - e\boldsymbol{A})\chi' \\ \left(i\hbar\frac{\partial}{\partial t} - e\varphi + 2mc^2\right) \chi' &= c\boldsymbol{\sigma} \cdot (\hat{\boldsymbol{p}} - e\boldsymbol{A})\phi' \end{aligned}} \tag{2.6}$$

In Gaussian units, the equations are

$$\begin{aligned} \left(i\hbar\frac{\partial}{\partial t} - e\varphi\right) \phi' &= c\boldsymbol{\sigma} \cdot \left(\hat{\boldsymbol{p}} - \frac{e}{c}\boldsymbol{A}\right) \chi' \\ \left(i\hbar\frac{\partial}{\partial t} - e\varphi + 2mc^2\right) \chi' &= c\boldsymbol{\sigma} \cdot \left(\hat{\boldsymbol{p}} - \frac{e}{c}\boldsymbol{A}\right) \phi' \end{aligned} \tag{2.7}$$

Since we will only deal with the primed quantities in the subsequent derivation, the primes will be omitted from hereon.

2.1 First-Order Approximation: Pauli Equation

In the first-order approximation, we only keep the $2mc^2$ term in the second equation of Eq. (2.6), because in non-relativistic limit, the particle's energy is much smaller than its rest energy. In this case,

$$\boxed{\chi^{(1)} \approx \frac{1}{2mc}\boldsymbol{\sigma}\cdot(\hat{\boldsymbol{p}} - e\boldsymbol{A})\phi} \tag{2.8}$$

Clearly,

$$\chi^{(1)} \sim \frac{1}{c}\phi \tag{2.9}$$

Substituting Eq. (2.8) into the first equation of Eq. (2.6), we have the equation for ϕ,

$$\left(i\hbar\frac{\partial}{\partial t} - e\varphi\right)\phi = \frac{1}{2m}[\boldsymbol{\sigma}\cdot(\hat{\boldsymbol{p}} - e\boldsymbol{A})]^2\phi \tag{2.10}$$

The Pauli matrices has the property

$$(\boldsymbol{\sigma}\cdot\boldsymbol{a})(\boldsymbol{\sigma}\cdot\boldsymbol{b}) = \boldsymbol{a}\cdot\boldsymbol{b} + i\boldsymbol{\sigma}\cdot(\boldsymbol{a}\times\boldsymbol{b}) \tag{2.11}$$

Thus

$$\begin{aligned}
[\boldsymbol{\sigma}\cdot(\hat{\boldsymbol{p}} - e\boldsymbol{A})]^2\phi &= (\hat{\boldsymbol{p}} - e\boldsymbol{A})^2\phi + i\boldsymbol{\sigma}\cdot[(\hat{\boldsymbol{p}} - e\boldsymbol{A})\times(\hat{\boldsymbol{p}} - e\boldsymbol{A})]\phi \\
&= (\hat{\boldsymbol{p}} - e\boldsymbol{A})^2\phi - ie\boldsymbol{\sigma}\cdot[\boldsymbol{A}\times(\hat{\boldsymbol{p}}\phi) + \hat{\boldsymbol{p}}\times(\boldsymbol{A}\phi)] \\
&= [(\hat{\boldsymbol{p}} - e\boldsymbol{A})^2 - ie\boldsymbol{\sigma}\cdot(\hat{\boldsymbol{p}}\times\boldsymbol{A})]\phi \\
&= [(\hat{\boldsymbol{p}} - e\boldsymbol{A})^2 - e\hbar\boldsymbol{\sigma}\cdot(\nabla\times\boldsymbol{A})]\phi \\
&= [(\hat{\boldsymbol{p}} - e\boldsymbol{A})^2 - e\hbar\boldsymbol{\sigma}\cdot\boldsymbol{B}]\phi
\end{aligned} \tag{2.12}$$

Therefore, the equation for ϕ becomes

$$\boxed{i\hbar\frac{\partial}{\partial t}\phi = \left[\frac{1}{2m}(\hat{\boldsymbol{p}} - e\boldsymbol{A})^2 + e\varphi - \frac{e\hbar}{2m}\boldsymbol{\sigma}\cdot\boldsymbol{B}\right]\phi} \tag{2.13}$$

This is the **Pauli equation**. It differs from the non-relativistic Schrödinger equation by the last term, which is **the potential energy of a magnetic dipole in an magnetic field.** Thus, in the first-order approximation with respect to $1/c$, the electron behaves as a particle with a charge and a magnetic dipole moment,

$$\boldsymbol{\mu} = \frac{e}{m}\boldsymbol{s} \tag{2.14}$$

where $\boldsymbol{s} = \frac{1}{2}\hbar\boldsymbol{\sigma}$ is the spin angular momentum. The ratio $\frac{e}{m}$ is the **gyromagnetic ratio** for electronic spin, which is **two times larger than the gyromagnetic ratio of orbital motion.**

The particle density is $\rho = \psi^*\psi = \phi^*\phi + \chi^*\chi$. In the first-order approximation, the last term should be dropped, because $\chi^{(1)*}\chi^{(1)} \sim 1/c^2$ is a higher-order term. The final result $\rho = |\phi|^2$ is as expected in non-relativistic quantum mechanics.

In SI units, the current density is given by

$$\boldsymbol{j} = c\psi^*\boldsymbol{\alpha}\psi = c(\phi^*\boldsymbol{\sigma}\chi + \chi^*\boldsymbol{\sigma}\phi) \tag{2.15}$$

Substituting the expression of $\chi^{(1)}$ given in Eq. (2.8), and using the properties of the Pauli matrices,

$$(\boldsymbol{\sigma} \cdot \boldsymbol{a})\boldsymbol{\sigma} = \boldsymbol{a} + i\boldsymbol{\sigma} \times \boldsymbol{a}, \quad \boldsymbol{\sigma}(\boldsymbol{\sigma} \cdot \boldsymbol{a}) = \boldsymbol{a} + i\boldsymbol{a} \times \boldsymbol{\sigma} \tag{2.16}$$

along with the formula,

$$\nabla \times (a\boldsymbol{A}) = (\nabla a) \times \boldsymbol{A} + a(\nabla \times \boldsymbol{A}) \tag{2.17}$$

we have

$$\boxed{\boldsymbol{j} = \frac{i\hbar}{2m}(\phi\nabla\phi^* - \phi^*\nabla\phi) - \frac{e}{m}\boldsymbol{A}\phi^*\phi + \frac{\hbar}{2m}\nabla \times (\phi^*\boldsymbol{\sigma}\phi)} \tag{2.18}$$

In Gaussian units, the Pauli equation is

$$i\hbar\frac{\partial}{\partial t}\phi = \left[\frac{1}{2m}\left(\hat{\boldsymbol{p}} - \frac{e}{c}\boldsymbol{A}\right)^2 + e\varphi - \frac{e\hbar}{2mc}\boldsymbol{\sigma} \cdot \boldsymbol{B}\right]\phi \tag{2.19}$$

which can be derived following the same procedure from Eq. (2.7). The gyromagnetic ratio in Gaussian units is $\frac{e}{mc}$,

$$\boldsymbol{\mu} = \frac{e}{mc}\boldsymbol{s} \tag{2.20}$$

And the current density is

$$\boldsymbol{j} = \frac{i\hbar}{2m}(\phi\nabla\phi^* - \phi^*\nabla\phi) - \frac{e}{mc}\boldsymbol{A}\phi^*\phi + \frac{\hbar}{2m}\nabla \times (\phi^*\boldsymbol{\sigma}\phi) \tag{2.21}$$

2.2 Second-Order Approximation: Spin-Orbit Coupling

If we continue our expansion to the order $1/c^2$, and **assume a pure electric field**, i.e., $\boldsymbol{A} = 0$, the particle density becomes

$$\rho = \phi^*\phi + \chi^{(1)*}\chi^{(1)} = \phi^*\phi + \frac{\hbar^2}{4m^2c^2}(\nabla\phi^* \cdot \boldsymbol{\sigma})(\boldsymbol{\sigma} \cdot \nabla\phi) \tag{2.22}$$

Note that we only need $\chi^{(1)}$ to calculate ρ because including the higher-order expansion of χ will result in terms in the order higher than $1/c^2$, which should be omitted. We see that when the $1/c^2$ terms are included in the expansion, the expression of ρ is not consistent with the expected result in non-relativistic quantum mechanics, where $\rho = |\phi|^2$. In this case, the approximated wave equation for ϕ does not correspond to a Schrödinger equation. Therefore, we need to construct a new two-component quantity ϕ_{Sch} and find the wave equation for ϕ_{Sch}.

ϕ_{Sch} should satisfy the condition

$$\int \phi_{\text{Sch}}^*\phi_{\text{Sch}}\mathrm{d}\boldsymbol{r} = \int \left[\phi^*\phi + \frac{\hbar^2}{4m^2c^2}(\nabla\phi^* \cdot \boldsymbol{\sigma})(\boldsymbol{\sigma} \cdot \nabla\phi)\right]\mathrm{d}\boldsymbol{r} \tag{2.23}$$

Using the identities

$$\nabla \cdot (a\boldsymbol{A}) = (\nabla a) \cdot \boldsymbol{A} + a(\nabla \cdot \boldsymbol{A}) \tag{2.24}$$

$$\nabla \cdot [\boldsymbol{A}(\nabla \cdot \boldsymbol{B})] = (\nabla \cdot \boldsymbol{A})(\nabla \cdot \boldsymbol{B}) + (\boldsymbol{A} \cdot \nabla)(\nabla \cdot \boldsymbol{B}) \tag{2.25}$$

we have

$$\begin{aligned}(\nabla\phi^*\cdot\boldsymbol{\sigma})(\boldsymbol{\sigma}\cdot\nabla\phi) &= [\nabla\cdot(\phi^*\boldsymbol{\sigma})][\nabla\cdot(\boldsymbol{\sigma}\phi)]\\ &= \nabla\cdot\{(\phi^*\boldsymbol{\sigma})[\nabla\cdot(\boldsymbol{\sigma}\phi)]\}-(\phi^*\boldsymbol{\sigma})\cdot\nabla[\nabla\cdot(\boldsymbol{\sigma}\phi)]\\ &= \nabla\cdot\{(\phi^*\boldsymbol{\sigma})[\nabla\cdot(\boldsymbol{\sigma}\phi)]\}-\phi^*(\boldsymbol{\sigma}\cdot\nabla)(\boldsymbol{\sigma}\cdot\nabla)\phi\\ &= \nabla\cdot\{(\phi^*\boldsymbol{\sigma})[\nabla\cdot(\boldsymbol{\sigma}\phi)]\}-\phi^*\nabla^2\phi \end{aligned}\tag{2.26}$$

Then integrating by parts,

$$\begin{aligned}\int(\nabla\phi^*\cdot\boldsymbol{\sigma})(\boldsymbol{\sigma}\cdot\nabla\phi)\mathrm{d}\boldsymbol{r} &= \int\nabla\cdot\{(\phi^*\boldsymbol{\sigma})[\nabla\cdot(\boldsymbol{\sigma}\phi)]\}\mathrm{d}\boldsymbol{r}-\int\phi^*\nabla^2\phi\mathrm{d}\boldsymbol{r}\\ &= \oint(\phi^*\boldsymbol{\sigma})[\nabla\cdot(\boldsymbol{\sigma}\phi)]\cdot\mathrm{d}\boldsymbol{S}-\int\phi^*\nabla^2\phi\mathrm{d}\boldsymbol{r}\end{aligned}\tag{2.27}$$

The first term equals to zero because for a bound state, ϕ vanishes at infinity. Alternatively, $(\nabla\phi^*\cdot\boldsymbol{\sigma})(\boldsymbol{\sigma}\cdot\nabla\phi)$ can be transformed to

$$\begin{aligned}(\nabla\phi^*\cdot\boldsymbol{\sigma})(\boldsymbol{\sigma}\cdot\nabla\phi) &= [\nabla\cdot(\phi^*\boldsymbol{\sigma})][\nabla\cdot(\boldsymbol{\sigma}\phi)]\\ &= \nabla\cdot\{[\nabla\cdot(\phi^*\boldsymbol{\sigma})](\boldsymbol{\sigma}\phi)\}-\nabla[\nabla\cdot(\phi^*\boldsymbol{\sigma})]\cdot(\boldsymbol{\sigma}\phi)\\ &= \nabla\cdot\{[\nabla\cdot(\phi^*\boldsymbol{\sigma})](\boldsymbol{\sigma}\phi)\}-(\phi\boldsymbol{\sigma})\cdot\nabla[\nabla\cdot(\boldsymbol{\sigma}\phi^*)]\\ &= \nabla\cdot\{[\nabla\cdot(\phi^*\boldsymbol{\sigma})](\boldsymbol{\sigma}\phi)\}-\phi(\boldsymbol{\sigma}\cdot\nabla)(\boldsymbol{\sigma}\cdot\nabla)\phi^*\\ &= \nabla\cdot\{[\nabla\cdot(\phi^*\boldsymbol{\sigma})](\boldsymbol{\sigma}\phi)\}-\phi\nabla^2\phi^*\end{aligned}\tag{2.28}$$

The third line is derived by transposing the scalar $\nabla[\nabla\cdot(\phi^*\boldsymbol{\sigma})]\cdot(\boldsymbol{\sigma}\phi)$. Finally, we have

$$\begin{aligned}\int(\nabla\phi^*\cdot\boldsymbol{\sigma})(\boldsymbol{\sigma}\cdot\nabla\phi)\mathrm{d}\boldsymbol{r} &= -\int\phi^*\nabla^2\phi\mathrm{d}\boldsymbol{r}=-\int\phi\nabla^2\phi^*\mathrm{d}\boldsymbol{r}\\ &= -\frac{1}{2}\int\left(\phi^*\nabla^2\phi+\phi\nabla^2\phi^*\right)\mathrm{d}\boldsymbol{r}\end{aligned}\tag{2.29}$$

and

$$\int\phi^*_{\mathrm{Sch}}\phi_{\mathrm{Sch}}\mathrm{d}\boldsymbol{r}=\int\left[\phi^*\phi-\frac{\hbar^2}{8m^2c^2}\left(\phi^*\nabla^2\phi+\phi\nabla^2\phi^*\right)\right]\mathrm{d}\boldsymbol{r}\tag{2.30}$$

We can therefore define

$$\boxed{\phi_{\mathrm{Sch}}=\left(1+\frac{\hat{\boldsymbol{p}}^2}{8m^2c^2}\right)\phi}\tag{2.31}$$

By calculating $\phi^*_{\mathrm{Sch}}\phi_{\mathrm{Sch}}$ and discard the terms in the order $1/c^4$, it is evident that Eq. (2.30) is satisfied. Conversely,

$$\boxed{\phi=\left(1-\frac{\hat{\boldsymbol{p}}^2}{8m^2c^2}\right)\phi_{\mathrm{Sch}}}\tag{2.32}$$

It's easy to show that

$$\left(1-\frac{\hat{\boldsymbol{p}}^2}{8m^2c^2}\right)\left(1+\frac{\hat{\boldsymbol{p}}^2}{8m^2c^2}\right)\phi=\phi-\frac{\hat{\boldsymbol{p}}^4}{64m^4c^4}\phi\approx\phi\tag{2.33}$$

where the term of order $1/c^4$ is omitted.

Next, we calculate the higher-order expansion of χ. For notational simplicity, we define $\hat{\varepsilon} = i\hbar\frac{\partial}{\partial t}$. In a pure electric field, Eq. (2.6) becomes

$$\begin{aligned}(\hat{\varepsilon} - e\varphi)\phi &= c(\boldsymbol{\sigma}\cdot\hat{\boldsymbol{p}})\chi \\ (\hat{\varepsilon} - e\varphi + 2mc^2)\chi &= c(\boldsymbol{\sigma}\cdot\hat{\boldsymbol{p}})\phi\end{aligned} \tag{2.34}$$

Note that without the presence of a magnetic field, this equation is the same in SI and Gaussian units. We can formally solve the second equation as

$$\chi = \frac{1}{\hat{\varepsilon} - e\varphi + 2mc^2} c(\boldsymbol{\sigma}\cdot\hat{\boldsymbol{p}})\phi = \frac{1}{2mc}\left(1 + \frac{\hat{\varepsilon} - e\varphi}{2mc^2}\right)^{-1}(\boldsymbol{\sigma}\cdot\hat{\boldsymbol{p}})\phi \tag{2.35}$$

Since in non-relativistic limit, the particle's energy is much smaller than its rest energy, we can apply the Taylor expansion to approximate $\left(1 + \frac{\hat{\varepsilon}-e\varphi}{2mc^2}\right)^{-1}$. Expanding up to the second term, we have

$$\boxed{\chi^{(2)} \approx \frac{1}{2mc}\left(1 - \frac{\hat{\varepsilon} - e\varphi}{2mc^2}\right)(\boldsymbol{\sigma}\cdot\hat{\boldsymbol{p}})\phi} \tag{2.36}$$

Substituting the expression of $\chi^{(2)}$ into the first equation in Eq. (2.34), and replacing ϕ by ϕ_{Sch}, we have the equation

$$(\hat{\varepsilon} - e\varphi)\left(1 - \frac{\hat{\boldsymbol{p}}^2}{8m^2c^2}\right)\phi_{\mathrm{Sch}} = \frac{1}{2m}(\boldsymbol{\sigma}\cdot\hat{\boldsymbol{p}})\left(1 - \frac{\hat{\varepsilon} - e\varphi}{2mc^2}\right)(\boldsymbol{\sigma}\cdot\hat{\boldsymbol{p}})\left(1 - \frac{\hat{\boldsymbol{p}}^2}{8m^2c^2}\right)\phi_{\mathrm{Sch}} \tag{2.37}$$

After some simple algebra, we arrive at

$$\begin{aligned}i\hbar\frac{\partial}{\partial t}\phi_{\mathrm{Sch}} = &\left(\frac{\hat{\boldsymbol{p}}^2}{2m} + e\varphi\right)\phi_{\mathrm{Sch}} - \frac{\hat{\boldsymbol{p}}^4}{16m^3c^2}\phi_{\mathrm{Sch}} - \frac{\hat{\boldsymbol{p}}^2}{8m^2c^2}\left(i\hbar\frac{\partial}{\partial t}\phi_{\mathrm{Sch}}\right) \\ &+ \frac{e}{4m^2c^2}(\boldsymbol{\sigma}\cdot\hat{\boldsymbol{p}})\varphi(\boldsymbol{\sigma}\cdot\hat{\boldsymbol{p}})\phi_{\mathrm{Sch}} - \frac{e}{8m^2c^2}\varphi\hat{\boldsymbol{p}}^2\phi_{\mathrm{Sch}}\end{aligned} \tag{2.38}$$

In this derivation, we have omitted the terms of order $1/c^4$. The right hand side of this equation also contains the time derivative of ϕ_{Sch}. We can eliminate this term by solving this equation iteratively, i.e., substituting

$$i\hbar\frac{\partial}{\partial t}\phi_{\mathrm{Sch}} = \left(\frac{\hat{\boldsymbol{p}}^2}{2m} + e\varphi\right)\phi_{\mathrm{Sch}} + O\left(\frac{1}{c^2}\right) \tag{2.39}$$

into the right hand side and discard all terms of order higher than $1/c^2$. We have

$$\begin{aligned}-\frac{\hat{\boldsymbol{p}}^2}{8m^2c^2}\left(i\hbar\frac{\partial}{\partial t}\phi_{\mathrm{Sch}}\right) &\approx -\frac{\hat{\boldsymbol{p}}^2}{8m^2c^2}\left(\frac{\hat{\boldsymbol{p}}^2}{2m} + e\varphi\right)\phi_{\mathrm{Sch}} \\ &= -\frac{\hat{\boldsymbol{p}}^4}{16m^3c^2}\phi_{\mathrm{Sch}} - \frac{e}{8m^2c^2}\hat{\boldsymbol{p}}^2(\varphi\phi_{\mathrm{Sch}})\end{aligned} \tag{2.40}$$

And the final wave equation for ϕ_{Sch} becomes

$$i\hbar\frac{\partial}{\partial t}\phi_{\mathrm{Sch}} = \left\{\frac{\hat{\boldsymbol{p}}^2}{2m} + e\varphi - \frac{\hat{\boldsymbol{p}}^4}{8m^3c^2} + \frac{e}{4m^2c^2}\left[(\boldsymbol{\sigma}\cdot\hat{\boldsymbol{p}})\varphi(\boldsymbol{\sigma}\cdot\hat{\boldsymbol{p}}) - \frac{1}{2}\left(\varphi\hat{\boldsymbol{p}}^2 + \hat{\boldsymbol{p}}^2\varphi\right)\right]\right\}\phi_{\mathrm{Sch}} \tag{2.41}$$

It has the same form as the Schrödinger equation, $i\hbar\frac{\partial}{\partial t}\phi_{\text{Sch}} = \hat{H}\phi_{\text{Sch}}$, with a Hamiltonian

$$\hat{H} = \frac{\hat{\boldsymbol{p}}^2}{2m} + e\varphi - \frac{\hat{\boldsymbol{p}}^4}{8m^3c^2} + \frac{e}{4m^2c^2}\left[(\boldsymbol{\sigma}\cdot\hat{\boldsymbol{p}})\varphi(\boldsymbol{\sigma}\cdot\hat{\boldsymbol{p}}) - \frac{1}{2}\left(\varphi\hat{\boldsymbol{p}}^2 + \hat{\boldsymbol{p}}^2\varphi\right)\right] \tag{2.42}$$

To further simplify the equations, we note that

$$\begin{aligned}
(\boldsymbol{\sigma}\cdot\hat{\boldsymbol{p}})\varphi(\boldsymbol{\sigma}\cdot\hat{\boldsymbol{p}})\phi_{\text{Sch}} &= -\hbar^2(\boldsymbol{\sigma}\cdot\nabla)\varphi(\boldsymbol{\sigma}\cdot\nabla\phi_{\text{Sch}}) \\
&= -\hbar^2[\boldsymbol{\sigma}\cdot(\nabla\varphi)](\boldsymbol{\sigma}\cdot\nabla)\phi_{\text{Sch}} - \hbar^2\varphi(\boldsymbol{\sigma}\cdot\nabla)(\boldsymbol{\sigma}\cdot\nabla)\phi_{\text{Sch}} \\
&= \hbar^2(\boldsymbol{\sigma}\cdot\boldsymbol{E})(\boldsymbol{\sigma}\cdot\nabla)\phi_{\text{Sch}} - \hbar^2\varphi\nabla^2\phi_{\text{Sch}} \\
&= \hbar^2\boldsymbol{E}\cdot\nabla\phi_{\text{Sch}} + i\hbar^2\boldsymbol{\sigma}\cdot(\boldsymbol{E}\times\nabla\phi_{\text{Sch}}) - \hbar^2\varphi\nabla^2\phi_{\text{Sch}} \\
&= i\hbar\boldsymbol{E}\cdot\hat{\boldsymbol{p}}\phi_{\text{Sch}} - \hbar\boldsymbol{\sigma}\cdot(\boldsymbol{E}\times\hat{\boldsymbol{p}})\phi_{\text{Sch}} + \varphi\hat{\boldsymbol{p}}^2\phi_{\text{Sch}}
\end{aligned} \tag{2.43}$$

$$\begin{aligned}
\varphi\hat{\boldsymbol{p}}^2\phi_{\text{Sch}} - \frac{1}{2}\left(\varphi\hat{\boldsymbol{p}}^2 + \hat{\boldsymbol{p}}^2\varphi\right)\phi_{\text{Sch}} &= -\frac{1}{2}\left(\hat{\boldsymbol{p}}^2\varphi - \varphi\hat{\boldsymbol{p}}^2\right)\phi_{\text{Sch}} \\
&= \frac{1}{2}\hbar^2\left[\nabla\cdot\nabla(\varphi\phi_{\text{Sch}}) - \varphi\nabla^2\phi_{\text{Sch}}\right] \\
&= \frac{1}{2}\hbar^2\phi_{\text{Sch}}\nabla\cdot(\nabla\varphi) + \hbar^2(\nabla\varphi)\cdot(\nabla\phi_{\text{Sch}}) \\
&= -\frac{1}{2}\hbar^2(\nabla\cdot\boldsymbol{E})\phi_{\text{Sch}} - i\hbar\boldsymbol{E}\cdot\hat{\boldsymbol{p}}\phi_{\text{Sch}}
\end{aligned} \tag{2.44}$$

where $\boldsymbol{E} = -\nabla\varphi$ is the electric field. Thus

$$\left[(\boldsymbol{\sigma}\cdot\hat{\boldsymbol{p}})\varphi(\boldsymbol{\sigma}\cdot\hat{\boldsymbol{p}}) - \frac{1}{2}\left(\varphi\hat{\boldsymbol{p}}^2 + \hat{\boldsymbol{p}}^2\varphi\right)\right]\phi_{\text{Sch}} = \left[-\hbar\boldsymbol{\sigma}\cdot(\boldsymbol{E}\times\hat{\boldsymbol{p}}) - \frac{1}{2}\hbar^2(\nabla\cdot\boldsymbol{E})\right]\phi_{\text{Sch}} \tag{2.45}$$

We obtain the final expression of the Hamiltonian

$$\boxed{\hat{H} = \frac{\hat{\boldsymbol{p}}^2}{2m} + e\varphi - \frac{\hat{\boldsymbol{p}}^4}{8m^3c^2} - \frac{e\hbar}{4m^2c^2}\boldsymbol{\sigma}\cdot(\boldsymbol{E}\times\hat{\boldsymbol{p}}) - \frac{e\hbar^2}{8m^2c^2}(\nabla\cdot\boldsymbol{E})} \tag{2.46}$$

The first two terms are the non-relativistic kinetic energy and potential energy of an electron in an external electric field, which is the same as in non-relativistic quantum mechanics. The last three terms are the relativistic correction to the Hamiltonian of order $1/c^2$. The first of the three is due to the **relativistic dependence of energy on momentum**,

$$\begin{aligned}
E &= c\sqrt{\boldsymbol{p}^2 + m^2c^2} - mc^2 \\
&= mc^2\left(1 + \frac{\boldsymbol{p}^2}{2m^2c^2} - \frac{\boldsymbol{p}^4}{8m^4c^4} + ...\right) - mc^2 \\
&= \frac{\boldsymbol{p}^2}{2m} - \frac{\boldsymbol{p}^4}{8m^3c^2} + ...
\end{aligned} \tag{2.47}$$

This term accounts for effect that **the mass of the electron increases with its speed**.

The second term is called the **spin-orbit interaction** or **spin-orbit coupling** (SOC), which describes the **interaction of a moving magnetic moment with an electric field.** In a classical description, the electron has an intrinsic magnetic moment

$$\boldsymbol{\mu} = \frac{e\hbar}{2m}\boldsymbol{\sigma} \tag{2.48}$$

Let $\boldsymbol{v} = \boldsymbol{p}/m$ be the velocity of the moving electron, then in its rest frame, it feels a magnetic field

$$\boldsymbol{B} = \frac{\boldsymbol{E} \times \boldsymbol{v}}{c^2} = \frac{\boldsymbol{E} \times \boldsymbol{p}}{mc^2} \tag{2.49}$$

The interaction energy is thus

$$-\boldsymbol{\mu} \cdot \boldsymbol{B} = -\frac{e\hbar}{2m^2c^2}\boldsymbol{\sigma} \cdot (\boldsymbol{E} \times \boldsymbol{p}) \tag{2.50}$$

which differs by a factor 1/2 from the correct spin-orbit interaction term. This factor is called the **Thomas half**, which is **due to the general requirement of relativistic invariance together with the particular properties of the electron as a spinor particle with the corresponding value of gyromagnetic ratio** (see next section).

A more familiar expression of SOC in chemistry is given in terms of the orbital and spin angular momentum, $\hat{\boldsymbol{l}}$ and $\hat{\boldsymbol{s}}$. Consider a centrally symmetric electric field,

$$\boldsymbol{E} = -\frac{\boldsymbol{r}}{r}\frac{\mathrm{d}\varphi}{\mathrm{d}r} \tag{2.51}$$

$$\hat{H}_{\mathrm{SOC}} = \frac{e\hbar}{4m^2c^2}\frac{1}{r}\frac{\mathrm{d}\varphi}{\mathrm{d}r}\boldsymbol{\sigma} \cdot (\boldsymbol{r} \times \hat{\boldsymbol{p}}) \tag{2.52}$$

Introducing the orbital and spin angular momentum operators,

$$\hat{\boldsymbol{l}} = \boldsymbol{r} \times \hat{\boldsymbol{p}}, \quad \hat{\boldsymbol{s}} = \frac{1}{2}\hbar\boldsymbol{\sigma} \tag{2.53}$$

and the potential energy $U = e\varphi$, we have

$$\boxed{\hat{H}_{\mathrm{SOC}} = \frac{1}{2m^2c^2}\frac{1}{r}\frac{\mathrm{d}U}{\mathrm{d}r}\hat{\boldsymbol{l}} \cdot \hat{\boldsymbol{s}}} \tag{2.54}$$

For the case of Coulomb potential created by the nucleus in an atom,

$$U = -\frac{Ze^2}{4\pi\varepsilon_0 r} \tag{2.55}$$

where Z is nuclear charge number, then

$$\boxed{\hat{H}_{\mathrm{SOC}} = \frac{Ze^2}{8\pi\varepsilon_0 m^2c^2r^3}\hat{\boldsymbol{l}} \cdot \hat{\boldsymbol{s}}} \tag{2.56}$$

The last term of the three is called the **Darwin term**, which comes from **the quantum fluctuating motion of the electron.** According to Maxwell equation, this term is nonzero only at the points where there are charges creating the external field (at the nucleus). For the case of electrons in an atom,

$$\nabla \cdot \boldsymbol{E} = -\frac{Ze}{\varepsilon_0}\delta(\boldsymbol{r}) \tag{2.57}$$

and

$$\boxed{\hat{H}_{\mathrm{Darwin}} = \frac{Ze^2\hbar^2}{8m^2c^2\varepsilon_0}\delta(\boldsymbol{r})} \tag{2.58}$$

In Gaussian units, the SOC and Darwin terms for the electron in an atom become

$$\hat{H}_{\mathrm{SOC}} = \frac{Ze^2}{2m^2c^2r^3}\hat{\boldsymbol{l}} \cdot \hat{\boldsymbol{s}}, \quad \hat{H}_{\mathrm{Darwin}} = \frac{\pi Ze^2\hbar^2}{2m^2c^2}\delta(\boldsymbol{r}) \tag{2.59}$$

2.3 Fine Structure of Hydrogen Atom

The $1/c^2$ terms in the Hamiltonian Eq. (2.46) give rise to relativistic corrections to the energy levels of a hydrogen atom. More generally, for a hydrogen-like atom with nuclear charge number Z, when $Z\alpha \ll 1$, these $1/c^2$ terms can be treated as a perturbation. Here

$$\boxed{\alpha = \frac{e^2}{4\pi\varepsilon_0 \hbar c}} \tag{2.60}$$

is the fine structure constant. $\alpha \approx 1/137$. Switch back to relativistic units, the perturbation operator $\hat{V}$ is

$$\hat{V} = -\frac{\hat{\boldsymbol{p}}^4}{8m^3} + \frac{Z\alpha}{2m^2 r^3}\hat{\boldsymbol{l}} \cdot \hat{\boldsymbol{s}} + \frac{Z\alpha\pi}{2m^2}\delta(\boldsymbol{r}) \tag{2.61}$$

The zeroth-order wave functions are solutions to the non-relativistic Schrödinger equation. Applying standard perturbation theory, the calculated correction to the energy levels is

$$\boxed{\Delta\varepsilon = -\frac{m(Z\alpha)^4}{2n^3}\left(\frac{1}{j+\frac{1}{2}} - \frac{3}{4n}\right)} \tag{2.62}$$

where n is the principal quantum number, j is the quantum number for total (orbital and spin) angular momentum.

In non-relativistic quantum mechanics, there are two degeneracies in a hydrogen-like atom. (1) degeneracy with respect to spin direction, and (2) degeneracy with respect to l for the same n. The relativistic correction (specifically, the SOC) removes the second degeneracy. **In Dirac theory, states with the same n and j are degenerate**. This degeneracy will be removed by radiative corrections (the Lamb shift).

3 Quasi-Classical Motion of a Spin in an External Field

In this section, we derive the quasi-classical equation of motion of a spin in an external EM field, and discuss the origin of the "Thomas half".

When the de Broglie wave length of the particle is much smaller than the characteristic length scale of the problem of interest, the properties of the system are very close to be classical. To formally derive the quasi-classical equation of motion, we make the following substitution to the second-order wave equation Eq. (1.31),

$$\psi = u e^{\frac{i}{\hbar}S} \tag{3.1}$$

where S is a scalar corresponding to the classical action, u is a slowly varying bispinor. Omitting the derivatives of u, we have a differential equation for the action S (in SI units),

$$\left\{\left[\frac{1}{c^2}\left(\frac{\partial S}{\partial t} + e\varphi\right)^2 - (\nabla S - e\boldsymbol{A})^2 - m^2c^2\right] - i\hbar\left[\frac{1}{c^2}\frac{\partial^2 S}{\partial t^2} - \nabla^2 S + ie\boldsymbol{\Sigma}\cdot\boldsymbol{B} + \frac{e}{c}\boldsymbol{\alpha}\cdot\boldsymbol{E}\right]\right\} u = 0 \tag{3.2}$$

We have used the Lorentz gauge where $\partial_\mu A^\mu = \nabla \cdot \boldsymbol{A} + \frac{1}{c^2}\frac{\partial \phi}{\partial t} = 0$. Next, we expand S in powers of $\hbar$ as

$$S = S_0 + \frac{\hbar}{i} S_1 + \left(\frac{\hbar}{i}\right)^2 S_2 + ... \tag{3.3}$$

If we only keep the zeroth-order term in $\hbar$, we have

$$\boxed{\frac{1}{c^2}\left(\frac{\partial S_0}{\partial t} + e\varphi\right)^2 - (\nabla S_0 - e\boldsymbol{A})^2 - m^2c^2 = 0} \tag{3.4}$$

This is the **classical relativistic Hamilton-Jacobi equation**. All terms which contain the spin are absent in the zeroth-order equation, and will only appear in the next approximation with respect to $\hbar$. Thus, **the influence of the magnetic moment of the electron on its motion is always of the same order of magnitude as the quantum corrections.** This is as expected, because spin is a purely quantum property and its magnitude is proportional to $\hbar$.

Here, instead of continuing the expansion of S with respect to $\hbar$, we adapt a different approach that does not directly involve the Dirac equation. The advantage is that the following treatment can be applied to any particles, including particles with "anomalous" gyromagnetic ratio that are not describable by the Dirac equation.

3.1 Non-Relativistic Case

The non-relativistic Hamiltonian of a spin $\frac{1}{2}$ particle with charge e and magnetic moment μ is given by

$$\hat{H} = \hat{H}' - \mu\boldsymbol{\sigma} \cdot \boldsymbol{B} \tag{3.5}$$

where in SI units,

$$\hat{H}' = \frac{1}{2m}(\hat{\boldsymbol{p}} - e\boldsymbol{A})^2 + e\varphi \tag{3.6}$$

includes all terms independent of spin. For Dirac particles (particles described by the Dirac equation), in SI units,

$$\mu = \frac{e\hbar}{2m} \tag{3.7}$$

For nucleons (protons and neutrons), μ also contains an "anomalous" part, μ',

$$\mu = \mu' + \frac{e\hbar}{2m} \tag{3.8}$$

The magnetic moment of electrons also contains an anomalous part if the radiative corrections are taken into account.

In Heisenberg's picture, the equation of motion of the spin operator $\hat{\boldsymbol{s}} = \frac{\hbar}{2}\boldsymbol{\sigma}$ is

$$\frac{\mathrm{d}}{\mathrm{d}t}\hat{\boldsymbol{s}} = \frac{i}{\hbar}[\hat{H}, \hat{\boldsymbol{s}}] = \frac{i}{2}(\hat{H}\boldsymbol{\sigma} - \boldsymbol{\sigma}\hat{H}) \tag{3.9}$$

Substituting the expression of $\hat{H}$ and noting that $\hat{H}'$ commute with $\boldsymbol{\sigma}$, we have

$$\frac{\mathrm{d}}{\mathrm{d}t}\hat{\boldsymbol{s}} = \frac{2\mu}{\hbar}\hat{\boldsymbol{s}} \times \boldsymbol{B} \tag{3.10}$$

where we have used the property of Pauli matrices in Eq. (2.16). For a quasi classical equation of motion, we average the operator equation over the states of the wave packet moving in a given path, i.e., we replace the operator $\hat{\boldsymbol{s}} = \frac{\hbar}{2}\boldsymbol{\sigma}$ by its expectation value $\frac{\hbar}{2}\boldsymbol{\zeta}$ ($\boldsymbol{\zeta}$ is called the **polarization vector**), and $\boldsymbol{B} = \boldsymbol{B}(t)$ represents the magnetic field at the position of the wave packet as it moves along the path. Thus the **non-relativistic quasi-classical equation of motion for the spin** is

$$\boxed{\frac{\mathrm{d}\boldsymbol{\zeta}}{\mathrm{d}t} = \frac{2\mu}{\hbar}\boldsymbol{\zeta} \times \boldsymbol{B}} \tag{3.11}$$

Clearly, the magnetic moment vector $\mu\boldsymbol{\zeta}$ precesses about the direction of the field with an angular velocity

$$\omega_{\boldsymbol{\zeta}} = -\frac{2\mu|\boldsymbol{B}|}{\hbar} \tag{3.12}$$

Now consider the orbital motion in non-relativistic case, the classical equation of motion of a charged particle in a magnetic field is

$$m\frac{\mathrm{d}\boldsymbol{v}}{\mathrm{d}t} = e\boldsymbol{v} \times \boldsymbol{B} \tag{3.13}$$

The particle's velocity rotates around the direction of the field with an angular velocity

$$\omega_{\boldsymbol{v}} = -\frac{e|\boldsymbol{B}|}{m} \tag{3.14}$$

For the case of a Dirac particle with no anomalous magnetic moment,

$$\omega_{\boldsymbol{\zeta}} = \omega_{\boldsymbol{v}} \tag{3.15}$$

The polarization vector is at a constant angle to the direction of motion. This conclusion is still valid in the relativistic case.

3.2 Relativistic Case

Next, we generalize Eq. (3.11) to the relativistic case. Relativistic units will be used from hereon. The non-relativistic formalism is applicable in the rest frame of the particle, where the polarization state of the particle is determined by the 3-vector $\boldsymbol{\zeta}$. Let $\boldsymbol{\zeta}$ be the space components of an axial 4-vector a^μ such that in the rest frame,

$$a^\mu = (0, \boldsymbol{\zeta}) \tag{3.16}$$

Then in a moving frame, the polarization of the particle will be given by the space component of a^μ, $\boldsymbol{a}$, in that frame. Note that a^μ is orthogonal to the 4-momentum $p^\mu = (\varepsilon, \boldsymbol{p})$,

$$\boxed{a^\mu p_\mu = 0} \tag{3.17}$$

This is trivial to prove by noting that $p^\mu = (m, \boldsymbol{0})$ in the rest frame. a^μ only has three independent components. Moreover, we have in any frame,

$$\boxed{a^\mu a_\mu = -\boldsymbol{\zeta}^2} \tag{3.18}$$

Let $\boldsymbol{v} = \boldsymbol{p}/\varepsilon$ be the velocity of the particle in a moving frame, by projecting $\boldsymbol{\zeta}$ on $\boldsymbol{v}$ and perform the Lorentz transformation, we have

$$a^0 = \frac{|\boldsymbol{p}|}{m}|\boldsymbol{\zeta}_\parallel|, \quad \boldsymbol{a}_\perp = \boldsymbol{\zeta}_\perp, \quad , \boldsymbol{a}_\parallel = \frac{\varepsilon}{m}\boldsymbol{\zeta}_\parallel \tag{3.19}$$

Or

$$\boxed{a^0 = \frac{\boldsymbol{\zeta}\cdot\boldsymbol{p}}{m}, \quad \boldsymbol{a} = \boldsymbol{\zeta} + \frac{(\boldsymbol{\zeta}\cdot\boldsymbol{p})\boldsymbol{p}}{m(\varepsilon+m)}} \tag{3.20}$$

This equation connects the polarization vector in any frame with that in an "instantaneous" rest frame.

The relativistic invariant equation of motion for the spin should be expressed in terms of the 4-vector a^μ and its derivative with respect to the proper time τ (τ is a 4-scalar). As a generalization of Eq. (3.11), relativistic invariance requires the right hand side be linear and homogeneous in the field tensor $F^{\mu\nu}$ and in the 4-vector a^μ. The only other quantity that can be included in the equation is the 4-velocity $u^\mu = p^\mu/m$. The only form of equation that satisfy these conditions is

$$\frac{\mathrm{d}a^\mu}{\mathrm{d}\tau} = AF^{\mu\nu}a_\nu + Bu^\mu F^{\nu\lambda}u_\nu a_\lambda \tag{3.21}$$

where A and B are constant coefficients. From the condition $a^\mu u_\mu = 0$ and the antisymmetry of $F^{\mu\nu}$ (whence $F^{\mu\nu}u_\mu u_\nu = 0$), no other expressions with the required properties can be constructed.

When $\boldsymbol{v} \to 0$, this equation must revert to Eq. (3.11). Setting $a^\mu = (0, \boldsymbol{\zeta})$, $u^\mu = (1, \mathbf{0})$, $\tau = t$, the differential equation for the space component becomes

$$\frac{\mathrm{d}\boldsymbol{\zeta}}{\mathrm{d}t} = A\boldsymbol{\zeta} \times \boldsymbol{B} \tag{3.22}$$

By comparing this equation with Eq. (3.11), we have $A = 2\mu$.

To determine B, we take the derivative of the relation $a^\mu u_\mu = 0$ with respect to τ,

$$u_\mu \frac{\mathrm{d}a^\mu}{\mathrm{d}\tau} = -a_\mu \frac{\mathrm{d}u^\mu}{\mathrm{d}\tau} = -\frac{e}{m}F^{\mu\nu}a_\mu u_\nu = \frac{e}{m}F^{\mu\nu}u_\mu a_\nu \tag{3.23}$$

where we have substituted the relativistic equation for the orbital motion of a charged particle,

$$\frac{\mathrm{d}u^\mu}{\mathrm{d}\tau} = \frac{e}{m}F^{\mu\nu}u_\nu \tag{3.24}$$

Meanwhile, we contract Eq. (3.21) with u_μ, and noting that $u_\mu u^\mu = 1$,

$$u_\mu \frac{\mathrm{d}a^\mu}{\mathrm{d}\tau} = (2\mu + B)F^{\mu\nu}u_\mu a_\nu \tag{3.25}$$

Therefore,

$$B = -2\left(\mu - \frac{e}{2m}\right) = -2\mu' \tag{3.26}$$

Thus the final **relativistic equation of motion for the spin** is

$$\boxed{\frac{\mathrm{d}a^\mu}{\mathrm{d}\tau} = 2\mu F^{\mu\nu}a_\nu - 2\mu' u^\mu F^{\nu\lambda}u_\nu a_\lambda} \tag{3.27}$$

From this equation, it's clear that

$$a_\mu \frac{\mathrm{d}a^\mu}{\mathrm{d}\tau} = 0 \tag{3.28}$$

Since $a^\mu a_\mu = -\boldsymbol{\zeta}^2$, **the magnitude of the polarization vector in each instantaneous rest frame remains constant during its motion.**

To describe how the direction of $\boldsymbol{\zeta}$ in each instantaneous rest frame changes, we first write the equation for the space component of a^μ (in any frame). Substituting $F^{\mu\nu} = (-\boldsymbol{E}, \boldsymbol{B})$, $u^\mu = \frac{\varepsilon}{m}(1, \boldsymbol{v})$, $\mathrm{d}\tau = \frac{m}{\varepsilon}\mathrm{d}t$, we have

$$\frac{\mathrm{d}\boldsymbol{a}}{\mathrm{d}t} = \frac{2\mu m}{\varepsilon} a^0 \boldsymbol{E} + \frac{2\mu m}{\varepsilon}(\boldsymbol{a} \times \boldsymbol{B}) - \frac{2\mu'\varepsilon}{m}(\boldsymbol{a} \cdot \boldsymbol{E})\boldsymbol{v} + \frac{2\mu'\varepsilon}{m} a^0 (\boldsymbol{v} \cdot \boldsymbol{E})\boldsymbol{v} + \frac{2\mu'\varepsilon}{m}[\boldsymbol{v} \cdot (\boldsymbol{a} \times \boldsymbol{B})]\boldsymbol{v} \tag{3.29}$$

Using the relation $a^\mu u_\mu = 0$, one can show that

$$a^0 = \boldsymbol{a} \cdot \boldsymbol{v} \tag{3.30}$$

The closed equation for $\boldsymbol{a}$ is thus

$$\frac{\mathrm{d}\boldsymbol{a}}{\mathrm{d}t} = \frac{2\mu m}{\varepsilon}(\boldsymbol{a} \cdot \boldsymbol{v})\boldsymbol{E} + \frac{2\mu m}{\varepsilon}(\boldsymbol{a} \times \boldsymbol{B}) - \frac{2\mu'\varepsilon}{m}(\boldsymbol{a} \cdot \boldsymbol{E})\boldsymbol{v} + \frac{2\mu'\varepsilon}{m}(\boldsymbol{a} \cdot \boldsymbol{v})(\boldsymbol{v} \cdot \boldsymbol{E})\boldsymbol{v} + \frac{2\mu'\varepsilon}{m}[\boldsymbol{v} \cdot (\boldsymbol{a} \times \boldsymbol{B})]\boldsymbol{v} \tag{3.31}$$

Next, we substitute Eq. (3.20), which relates $\boldsymbol{a}$ in an arbitrary frame to $\boldsymbol{\zeta}$ in the instantaneous rest frame, into the equation above. Note that

$$\frac{\mathrm{d}\boldsymbol{p}}{\mathrm{d}t} = e\boldsymbol{E} + e\boldsymbol{v} \times \boldsymbol{B}, \quad \frac{\mathrm{d}\varepsilon}{\mathrm{d}t} = e\boldsymbol{v} \cdot \boldsymbol{E} \tag{3.32}$$

After a lengthy calculation, we have

$$\boxed{\frac{\mathrm{d}\boldsymbol{\zeta}}{\mathrm{d}t} = \frac{2\mu m + 2\mu'(\varepsilon - m)}{\varepsilon}(\boldsymbol{\zeta} \times \boldsymbol{B}) + \frac{2\mu'\varepsilon}{\varepsilon + m}(\boldsymbol{v} \cdot \boldsymbol{B})(\boldsymbol{v} \times \boldsymbol{\zeta}) + \frac{2\mu m + 2\mu'\varepsilon}{\varepsilon + m}[\boldsymbol{\zeta} \times (\boldsymbol{E} \times \boldsymbol{v})]} \tag{3.33}$$

Consider the **motion of a spin in a pure electric field at the low speed limit**, i.e., $\boldsymbol{B} = 0$, $\varepsilon \to m$, we have

$$\frac{\mathrm{d}\boldsymbol{\zeta}}{\mathrm{d}t} = (\mu + \mu')[\boldsymbol{\zeta} \times (\boldsymbol{E} \times \boldsymbol{v})] \tag{3.34}$$

Or

$$\boxed{\frac{\mathrm{d}\boldsymbol{\zeta}}{\mathrm{d}t} = \left(\frac{e}{2m} + 2\mu'\right)[\boldsymbol{\zeta} \times (\boldsymbol{E} \times \boldsymbol{v})]} \tag{3.35}$$

Note the additional factor of two in front of the anomalous magnetic moment μ' compared to the "normal" magnetic moment $\frac{e}{2m}$. If we impose the condition that this equation should be derived quantum mechanically, we need to find a Hamiltonian operator that gives the following equation of motion of the spin operator in Heisenberg picture:

$$\frac{\mathrm{d}\hat{\boldsymbol{s}}}{\mathrm{d}t} = \left(\frac{e}{2m} + 2\mu'\right)\frac{[\hat{\boldsymbol{s}} \times (\boldsymbol{E} \times \hat{\boldsymbol{p}})]}{m} \tag{3.36}$$

Using Eq. (2.16), one can easily find that

$$\boxed{\hat{H} = \hat{H}' - \left(\mu' + \frac{e}{4m}\right)\frac{[\boldsymbol{\sigma} \cdot (\boldsymbol{E} \times \hat{\boldsymbol{p}})]}{m}} \tag{3.37}$$

where $\hat{H}'$ contains the terms that do not explicitly depend on the spin. Now there is an additional factor of 1/2 in front of the "normal" magnetic moment $\frac{e}{2m}$ compared to the anomalous magnetic moment μ'. This is the Thomas half. It is clear that it originates from the requirement of relativistic invariance: Eq. (3.27) must has that specific form.

If there is no anomalous magnetic moment, $\mu' = 0$, the last term in Eq. (3.37) returns to the SOC term derived from Dirac equation. The other two $1/c^2$ terms will be included in $\hat{H}'$ because they do not explicitly depend on the particle's spin.